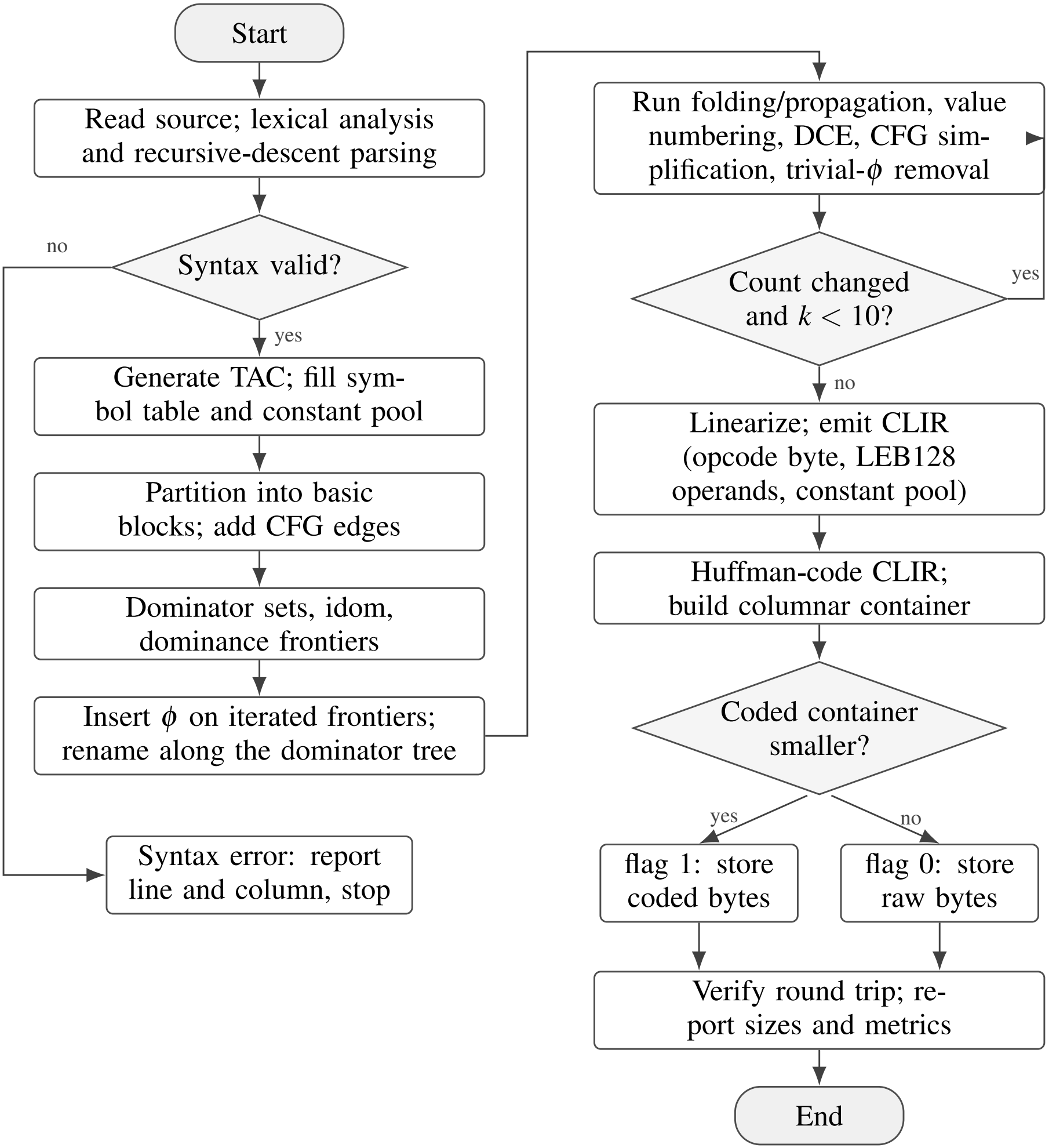
Start
Read source; lexical analysis and recursive-descent parsing
Syntax valid?
no
yes
Syntax error: report line and column, stop
Generate TAC; fill symbol table and constant pool
Partition into basic blocks; add CFG edges
Dominator sets, idom, dominance frontiers
Insert ϕ on iterated frontiers; rename along the dominator tree
Run folding/propagation, value numbering, DCE, CFG simplification, trivial-ϕ removal
Count changed and $k < 10$?
yes
no
Linearize; emit CLIR (opcode byte, LEB128 operands, constant pool)
Huffman-code CLIR; build columnar container
Coded container smaller?
yes
no
flag 1: store coded bytes
flag 0: store raw bytes
Verify round trip; report sizes and metrics
End